

Predict Hotel Booking Cancellation using Big Data Analytics*

*Department of Computer Science
Princess Nourah bint Abdulrahman University*

Abstract—Hotel booking cancellations are a major challenge in the hospitality industry because they affect revenue, room availability, and resource planning. This project uses the Hotel Booking Demand Dataset to analyze booking behavior and predict whether a reservation will be canceled. Exploratory Data Analysis was performed using PySpark to identify important patterns, trends, and outliers related to cancellations. Machine learning models were then built using PySpark MLlib, including Logistic Regression, Decision Tree, and Random Forest. The models were evaluated using accuracy, precision, recall, F1-score, and confusion matrices. The results show that big data analytics and machine learning can support hotels in improving cancellation prediction, demand forecasting, and decision-making.

I. INTRODUCTION

Hotel booking cancellations represent a critical issue in the hospitality industry, leading to significant revenue loss and inefficient resource utilization. With the rapid growth of digital booking platforms, the volume of reservation data has increased dramatically, creating opportunities to leverage Big Data Analytics for more accurate and scalable prediction systems.

This study aims to:

- Analyze large-scale hotel booking data to identify key factors influencing cancellations
- Develop predictive models for cancellation detection
- Evaluate the effectiveness of machine learning techniques in improving forecasting accuracy
- Explore the role of big data frameworks in handling large-scale, real-time prediction scenarios

A. Problem Statement

High hotel booking cancellation rates represent a significant challenge for the hospitality and tourism industry. Hotels rely heavily on reservation systems to manage room availability, staffing, and resource allocation. However, a large proportion of reservations are often canceled before the arrival date, which disrupts operational planning and leads to inefficient utilization of hotel resources. It also results in substantial financial losses, inaccurate demand forecasting, and difficulties in managing room inventory effectively.

With the availability of large volumes of booking data, data analytics and machine learning techniques can be utilized to analyze booking patterns and identify factors associated with cancellations. Such predictive approaches can help hotels

anticipate potential cancellations and improve decision-making processes.

B. Research Question

How accurately can machine learning models predict hotel booking cancellations based on historical reservation data and customer booking attributes?

C. SDG relevance

SDG 8: Decent Work and Economic Growth

This project supports SDG 8 by improving operational efficiency in the hospitality industry. By predicting bookings likely to be canceled using big data analytics, it helps hotels reduce financial losses, improve productivity, and support economic growth in the tourism sector.

II. DATASET DESCRIPTION

The study employs the Hotel Booking Demand Dataset obtained from Kaggle. The dataset contains booking information for both a resort hotel and a city hotel and is widely used for analyzing booking behavior and predicting cancellations.

A. Dataset Name

Hotel Booking Demand Dataset

B. Size

The dataset includes 119,390 booking records and 31 features.

C. Features

The dataset includes features describing customer behavior, reservation details, and booking outcomes. The main features include:

- Arrival date
- Lead time (days)
- Average Daily Rate (ADR) (price in USD)
- Customer type
- Deposit type
- Number of adults, children, and babies
- Number of special requests

D. Structure

The dataset is structured in tabular format, where each row represents a booking and each column represents a feature. It contains both completed and canceled reservations from July 2015 to August 2017.

The dataset is suitable for classification tasks, where the goal is to determine whether a booking will be canceled or not based on the target variable.

E. Units

- Price is measured in USD
- Lead time and duration-related variables are measured in days

III. BIG DATA ANALYTICS

Big Data Analytics refers to the process of analyzing large and complex datasets that cannot be efficiently processed using traditional data processing methods. In the context of hotel booking systems, the increasing volume of reservation data generated from online platforms requires scalable and efficient data processing frameworks.

Technologies such as Apache Spark and Hadoop enable distributed data processing, allowing large-scale datasets to be analyzed in parallel. These frameworks support faster computation, real-time data processing, and improved scalability compared to traditional systems.

In this study, Big Data Analytics plays a key role in handling large volumes of hotel booking data and improving the performance of predictive models. By leveraging scalable data processing techniques, it becomes possible to build more accurate and efficient models for predicting hotel booking cancellations.

IV. RELATED WORK

1) *Hotel Booking Demand Dataset*: A dataset called Hotel Booking Demand was published by António, Almeida, and Nunes (2019). Due to its comprehensive structure and real-world relevance, it is one of the most widely used datasets in hotel booking analysis and cancellation prediction research.

The dataset contains booking information for both a resort hotel and a city hotel between July 2015 and August 2017. It includes 119,390 booking records and 31 variables describing customer characteristics, reservation details, and booking outcomes. These variables include attributes such as lead time, number of special requests, deposit type, and customer type [1].

2) *Predicting Hotel Booking Cancellation Using Data Science*: Antonio, Almeida, and Nunes (2017) conducted a study that applied data science techniques to predict hotel booking cancellations.

Their research showed that predictive models can help hotels anticipate cancellations before the arrival date by analyzing historical reservation data. By identifying patterns in customer booking behavior, hotels can improve demand forecasting, optimize pricing strategies, and manage overbooking more effectively. The study emphasizes the role of machine

learning and data analytics in supporting decision-making in the hospitality industry [2].

However, this study is limited to traditional data processing techniques and does not address scalability or real-time prediction using big data frameworks.

3) *Predicting Hotel Booking Cancellations Using a Hybrid Tree-Based Neural Network Approach*: Several studies have applied machine learning techniques to predict hotel booking cancellations. Yang developed a Tree-Based Neural Network (TNN) model that combines tree-based algorithms with neural networks. The model was trained on historical booking data to analyze customer behavior and identify patterns related to cancellations. The proposed approach demonstrated better performance compared to traditional models. The results indicate that the proposed model improves prediction accuracy and supports more effective decision-making in booking and revenue management [3].

However, this approach is primarily based on traditional data processing methods and does not address scalability challenges associated with large-scale data environments.

4) *Machine Learning Models for Hotel Booking Cancellation Prediction*: Several studies have explored the use of machine learning techniques to predict hotel booking cancellations. Common classification algorithms such as Logistic Regression, Decision Trees, and Random Forest are widely used to analyze customer booking behavior based on historical reservation data. These approaches aim to identify patterns and relationships between different variables that influence whether a booking will be canceled.

Key features typically include lead time, previous cancellations, booking changes, and customer type. Among various methods, ensemble techniques such as Random Forest have shown strong performance due to their ability to handle complex interactions between variables and reduce overfitting. In addition, proper data preprocessing and feature engineering play an essential role in improving model accuracy.

Results from previous studies demonstrate that machine learning models effectively predict cancellation patterns and support decision-making in the hospitality sector. These insights enable hotels to optimize pricing strategies, improve demand forecasting, and manage overbooking more efficiently. This aligns with the objectives of this study, where similar techniques are applied to analyze hotel booking data and predict potential cancellations [4].

However, these studies mainly rely on traditional machine learning models and do not fully address scalability challenges in big data environments.

5) *Predicting Hotel Booking Cancellations Using Machine Learning and Big Data*: This study focuses on predicting hotel booking cancellations using machine learning and big data techniques. It utilizes real booking data to identify patterns associated with customer behavior and cancellation likelihood. The research applies artificial intelligence models, including neural networks optimized using genetic algorithms, to improve prediction accuracy and model performance. The findings demonstrate that such approaches can effectively identify

customers likely to cancel their bookings, enabling hotels to enhance decision-making, optimize revenue management, and reduce financial losses. From a Big Data perspective, the study highlights how large-scale booking data can be processed and analyzed efficiently to support accurate and scalable prediction systems [5].

While previous studies have demonstrated the effectiveness of machine learning models such as Random Forest, Logistic Regression, and hybrid neural networks in predicting hotel booking cancellations, these approaches are often implemented on relatively small-scale datasets and traditional processing environments.

However, fewer studies explore scalable big data frameworks for real-time prediction. In addition, there is limited comparison between models in terms of scalability, computational efficiency, and real-time applicability. This highlights a research gap in integrating Big Data technologies with predictive analytics to improve performance on large-scale hotel booking datasets.

V. EXPLORATORY DATA ANALYSIS

This section presents the exploratory data analysis conducted using PySpark. The analysis focuses on identifying patterns, trends, outliers, and key insights from the hotel booking dataset.

A. Dataset Overview

The dataset was loaded and analyzed using PySpark. Dataset contains 119,390 booking records and 31 features related to hotel reservations, customer behavior, and booking outcomes.

The analysis began by examining the dataset structure, column names, and data types using PySpark functions such as `printSchema()` and `describe()`. The dataset includes both numerical and categorical variables describing reservation details and customer information.

B. Data Cleaning and Validation

The dataset was validated and cleaned using PySpark before performing exploratory data analysis and machine learning tasks. PySpark functions such as `printSchema()` and `describe()` were used to examine the dataset structure, data types, and statistical summaries of numerical variables.

The analysis confirmed that the dataset did not contain missing values in the analyzed features. However, duplicate records were identified within the dataset. A total of 31,994 duplicate rows were detected and removed to improve the reliability and consistency of the analysis.

After removing duplicate records, the final dataset contained 87,396 rows. In addition, descriptive statistics revealed variations in variables such as `lead_time` and `adr`, with some extreme values indicating the presence of outliers.

C. Target Variable Distribution

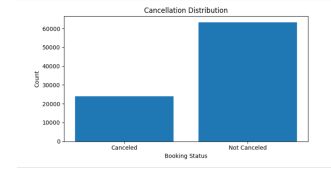


Fig. 1. cancellation distribution

The target variable `is_canceled` was analyzed to examine the distribution of canceled and non-canceled bookings after the data cleaning process.

The analysis showed that non-canceled bookings represented the majority of the dataset, while canceled bookings accounted for a large proportion of reservations. This indicates that hotel booking cancellation prediction is an important classification problem in the hospitality industry.

The distribution of the target variable also highlights the importance of evaluating machine learning models carefully to ensure reliable cancellation prediction performance.

D. Feature Analysis

Several important variables were analyzed to understand their relationship with hotel booking cancellations. Features such as `lead_time`, `adr`, `customer_type`, `market_segment`, and `deposit_type` showed strong relationships with cancellation behavior.

The analysis revealed that bookings with longer lead times were more likely to be canceled. In addition, Online Travel Agency bookings showed higher cancellation rates compared to direct and corporate bookings.

Customer type also influenced booking behavior, where transient customers represented the largest proportion of cancellations. Furthermore, bookings with no deposit were canceled more frequently than bookings with non-refundable deposits.

E. Patterns

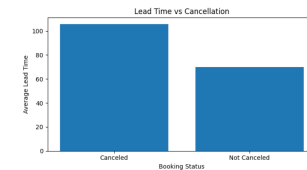


Fig. 2. Lead Time vs Cancellation

Pattern: Bookings with longer lead times exhibit higher cancellation rates, indicating that reservations made far in advance are more likely to be canceled.

TABLE I
SUMMARY OF RELATED WORK

Paper	Data	Target	Preprocessing	Model	Results
Hotel Booking Demand Dataset [1]	119,390 hotel booking records with 31 features	Understanding booking behavior	Data cleaning and feature selection	Statistical analysis	Widely used dataset for booking and cancellation prediction
Predicting Booking Cancellation [2]	Historical reservation data	Predicting cancellations	Data preprocessing and pattern analysis	Machine learning models	Improved forecasting and decision-making
Hybrid TNN Model [3]	Historical booking data	Predicting cancellations	Data preparation and feature extraction	Tree-based Neural Network	Higher accuracy than traditional models
ML Models for Cancellation [4]	Booking datasets	Classification of cancellations	Feature engineering and pre-processing	Random Forest, Decision Trees, Logistic Regression	Effective prediction with high performance
Hotel Booking Cancellation Prediction using Big Data [5]	Real booking data	Predicting cancellations	Data preprocessing and feature analysis	Neural Networks with Genetic Algorithms	High prediction accuracy and improved decision-making



Fig. 3. Customer Type vs Cancellation

Pattern: Transient customers represent the largest booking segment and show the highest cancellation counts, while Contract and Group customers have significantly lower cancellation behavior.

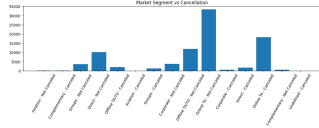


Fig. 4. Market Segment vs Cancellation

Pattern: Online Travel Agency (Online TA) and Offline TA/TO segments generate the highest booking and cancellation volumes, indicating that booking channels strongly influence customer cancellation behavior.

F. Outlier Detection

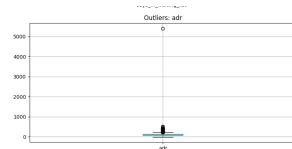


Fig. 5. ADR-outliers

Outlier: ADR shows the most significant outliers, with several bookings having exceptionally high room prices compared to the normal pricing range.

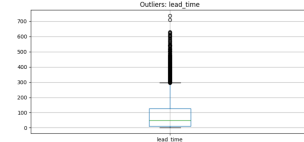


Fig. 6. leadtime-outliers

Outlier: Lead time reveals clear outliers where some reservations were made far earlier than the majority of bookings, indicating unusually long advance booking periods.

G. Trends

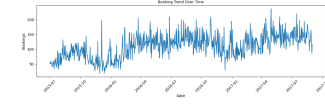


Fig. 7. Booking Trend Over Time

Trend: Hotel bookings vary across months, showing clear seasonal behavior. Higher booking volumes are observed during summer and holiday periods, reflecting increased travel demand. Cancellation rates also rise during peak seasons, indicating that customers are more likely to cancel reservations made during busy periods.

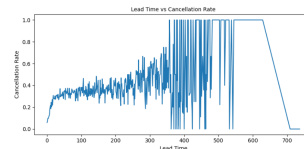


Fig. 8. Lead Time vs Cancellation Rate trend

Trend: Lead time shows a strong relationship with cancellations. Bookings made far in advance have a significantly higher probability of being canceled compared to last-minute bookings. This indicates that uncertainty increases over time, making early reservations less reliable.

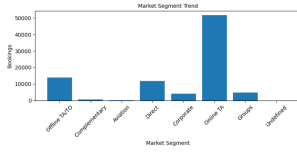


Fig. 9. Market Segment Trend

Trend: Online Travel Agencies (Online TA) dominate booking trends, contributing the largest share of reservations. However, they also show the highest cancellation rates. Direct bookings and corporate segments demonstrate more stable behavior with lower cancellation trends.

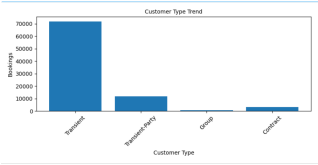


Fig. 10. Customer Type Trend

Trend: Transient customers represent the majority of bookings and show the highest variability in cancellation behavior. In contrast, contract and group customers demonstrate more consistent booking patterns with significantly lower cancellation rates.

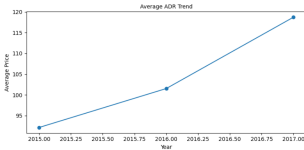


Fig. 11. Average ADR

Trend: Average Daily Rate (ADR) changes across seasons according to demand levels. Higher ADR values are observed during peak travel periods, reflecting dynamic pricing strategies. Some extremely high ADR values may indicate luxury bookings or unusual pricing outliers.

Summary: Overall, the results show that time-related variables, booking channels, and customer types strongly influence booking demand and cancellation behavior. These trends provide deeper insights into booking behavior and support the interpretation of key findings in the next section.

Key Insights The exploratory data analysis reveals several important insights into hotel booking cancellation behavior. One of the most significant findings is that lead time strongly influences cancellations, as bookings made far in advance are more likely to be canceled due to increased uncertainty over time. Not to mention that customer type plays a crucial role, with transient customers showing the highest cancellation

rates, while contract and group customers demonstrate more stable booking behavior.

The analysis also highlights that booking channels significantly impact cancellation patterns, where Online Travel Agencies generate the highest number of bookings as well as cancellations, whereas direct and corporate bookings are more stable. Furthermore, seasonal trends affect both booking demand and cancellations, with higher booking volumes and increased cancellation rates observed during peak travel periods such as summer and holidays.

From a Big Data perspective, the use of scalable technologies such as PySpark enables efficient processing of large datasets and supports the development of accurate predictive models. These insights demonstrate that predictive analytics can help hotels anticipate cancellations, optimize pricing strategies, improve demand forecasting, and reduce financial losses, ultimately enhancing decision-making in the hospitality industry

H. Correlation

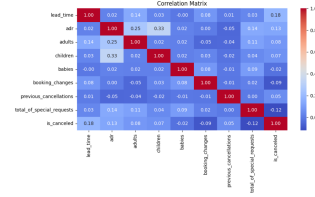


Fig. 12. Correlation

Correlation Analysis: The correlation matrix shows that lead_time has the highest positive correlation with is_canceled, indicating that bookings made earlier are more likely to be canceled. In addition, adr has a slight positive relationship with cancellations, while total_of_special_requests shows a negative relationship, suggesting that customers with more special requests are less likely to cancel their bookings.

VI. MODELS BUILDING

Here we will show what machine learning models developed using PySpark MLlib to predict hotel booking cancellations. Different classification algorithms were implemented and evaluated to identify the most effective model for cancellation prediction.

The machine learning workflow included data preprocessing, feature engineering, model training, and performance evaluation using several classification metrics.

A. Feature Engineering and Preprocessing

Before training the machine learning models, several preprocessing steps were performed using PySpark MLlib. Selected numerical features related to booking behavior and customer information were extracted from the dataset.

The preprocessing process included replacing invalid values, converting feature columns into numerical data types using

cast(), and removing rows containing null values to ensure data quality and consistency.

The selected features were then combined using VectorAssembler to create a unified feature vector for machine learning models. The dataset was divided into training and testing sets to evaluate model performance and generalization capability.

B. Logistic Regression

Logistic Regression was implemented using PySpark MLlib as a baseline classification model for predicting hotel booking cancellations. The model estimates the probability of a booking being canceled based on the input features extracted from the dataset.

Logistic Regression is widely used for binary classification problems because of its simplicity, interpretability, and computational efficiency. The model was trained using the prepared feature vectors and evaluated using classification performance metrics.

C. Decision Tree Classifier

A Decision Tree Classifier was developed using PySpark MLlib to analyze booking cancellation behavior through hierarchical decision rules. The model splits the dataset into branches based on feature importance and decision thresholds.

Decision Trees are effective for handling both numerical and categorical data and provide interpretable classification results. The model was trained on the processed dataset and used to identify important factors influencing hotel booking cancellations.

D. Random Forest Classifier

Random Forest was implemented using PySpark MLlib as an ensemble learning model for hotel booking cancellation prediction. The model combines multiple decision trees to improve classification accuracy and reduce overfitting.

Random Forest is considered one of the most effective classification algorithms for handling complex datasets because it improves prediction stability and generalization performance. The model was trained and evaluated using the prepared training and testing datasets.

VII. EXPERIMENTS SECTION

A. Train-Test Split

The dataset was divided into training and testing sets to evaluate model performance. The training dataset was used to train the models, while the testing dataset was used to measure prediction accuracy and generalization performance.

B. Performance Evaluation

The machine learning models were evaluated using several classification performance metrics to measure their prediction effectiveness on hotel booking cancellations.

Evaluation metrics included Accuracy, Precision, Recall, and F1-score. These metrics were calculated using PySpark MLlib evaluators on the testing dataset.

Accuracy measures the overall prediction correctness of the model, while Precision evaluates the proportion of correctly predicted cancellation cases. Recall measures the model's ability to identify actual canceled bookings, and F1-score provides a balanced evaluation between Precision and Recall.

The evaluation results were compared across Logistic Regression, Decision Tree, and Random Forest models to identify the best-performing algorithm for hotel booking cancellation prediction.

C. Experiments and Results

The machine learning models were trained and evaluated using the prepared training and testing datasets. Logistic Regression, Decision Tree, and Random Forest models were compared using Accuracy, Precision, Recall, and F1-score metrics.

TABLE II
MODEL PERFORMANCE COMPARISON

Model	Accuracy	Precision	Recall	F1-score
Logistic Regression	0.7381	0.7068	0.7381	0.6731
Decision Tree	0.7609	0.7399	0.7609	0.7351
Random Forest	0.7487	0.7464	0.7487	0.6776

Results Discussion: The experimental results show that the Decision Tree model achieved the highest overall performance among the implemented models, with an accuracy of 0.7609 and an F1-score of 0.7351. Logistic Regression achieved competitive performance as a baseline model, while Random Forest also produced strong results with high precision values.

D. Cross Validation Comparison

TABLE III
COMPARISON BETWEEN MODELS WITH AND WITHOUT CROSS VALIDATION

Model	Accuracy (No CV)	Accuracy (CV)	F1 (No CV)	F1 (CV)
Logistic Regression	0.7400	0.7381	0.6814	0.6731
Decision Tree	0.7590	0.7609	0.7355	0.7351
Random Forest	0.7487	0.7487	0.6776	0.6776

Cross Validation Analysis: The comparison between models trained with and without Cross Validation (CV) shows that the overall performance remained relatively stable across all models. Decision Tree achieved the best overall performance in both experiments, while Random Forest maintained consistent results.

E. Confusion Matrix Analysis

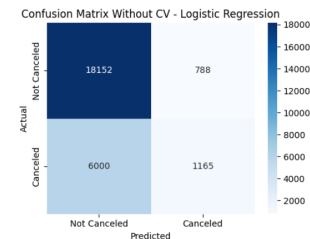


Fig. 13. Confusion Matrix for Logistic Regression Without Cross Validation

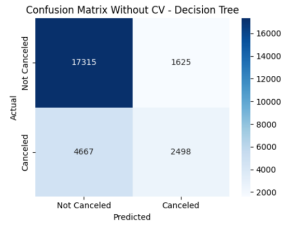


Fig. 14. Confusion Matrix for Decision Tree Without Cross Validation

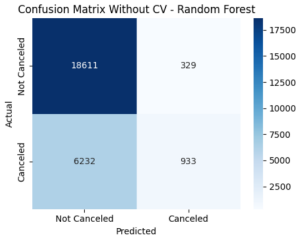


Fig. 15. Confusion Matrix for Random Forest Without Cross Validation

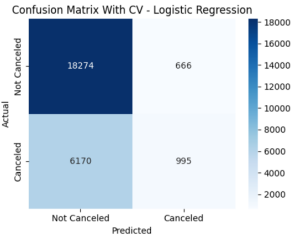


Fig. 16. Confusion Matrix for Logistic Regression With Cross Validation

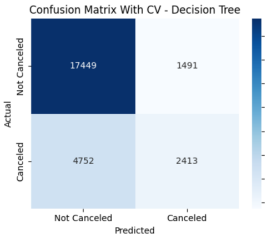


Fig. 17. Confusion Matrix for Decision Tree With Cross Validation

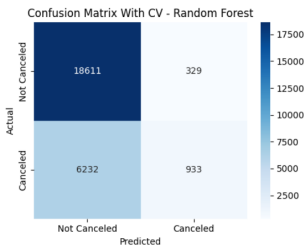


Fig. 18. Confusion Matrix for Random Forest With Cross Validation

Confusion Matrix Analysis: The confusion matrices provide a detailed evaluation of the classification performance for each machine learning model. The results show that Decision Tree achieved better balance between correctly predicted canceled and non-canceled bookings compared to the other models. Logistic Regression and Random Forest produced strong performance in predicting non-canceled bookings, while Decision Tree showed improved capability in identifying canceled reservations.

VIII. CONCLUSION

This project explored the use of Big Data Analytics and machine learning techniques to predict hotel booking cancellations using the Hotel Booking Demand Dataset. Exploratory Data Analysis was conducted using PySpark to identify important patterns, trends, outliers, and factors influencing cancellation behavior.

Several machine learning models were implemented using PySpark MLlib, including Logistic Regression, Decision Tree, and Random Forest. The models were evaluated using Accuracy, Precision, Recall, F1-score, and confusion matrices to compare their prediction performance.

The experimental results showed that the Decision Tree model achieved the best overall performance among the implemented models. In addition, the use of PySpark enabled efficient processing and analysis of large-scale booking data.

Overall, the results demonstrate that machine learning and big data technologies can effectively support hotel management in predicting booking cancellations, improving demand forecasting, reducing financial losses, and enhancing decision-making processes.

IX. REFERENCES

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